Aayush Patel

Atlanta, GA | (770) 862-5673 | apatel3088@gatech.edu | U.S. Permanent Resident, no sponsorship required
linkedin.com/in/aysh-ptl | github.com/AayushP-418 | aayush-patel-portfolio-tezw.vercel.app

Education

Georgia Institute of Technology Atlanta, GA
B.S. Computer Science, Intelligence and Systems Architecture; Minor: FinTech *Expected May 2028*

- GPA: 3.75 / 4.00. Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Objects & Design
- Honors: Faculty Honors; Dean's List

Technical Skills

Languages: Python, SQL, Java, TypeScript, Go, JavaScript
AI & ML: PyTorch, Hugging Face, Transformers, QLoRA/SFT, scikit-learn, XGBoost, RAG, LLM Evaluation, Tool Calling
Data & APIs: FastAPI, Spring Boot, PostgreSQL, pgvector/HNSW, Redis, ChromaDB, SQLAlchemy, Pydantic, REST
Cloud & Engineering: AWS (ECS Fargate, RDS, ElastiCache, SQS, Lambda), GCP, Docker, Terraform, CI/CD, Linux

Experience

Pinnacle Compounding Pharmacy Aug 2026 - Present
Data Science Intern *Phoenix, AZ (Remote)*

- Building Python data services over claims and remittance data on a governed PostgreSQL schema for downstream analytics
- Shipping an LLM extraction pipeline parsing 2K denial letters into structured records at 92% field-level accuracy
- Implemented pgvector/HNSW payer-policy retrieval with expert-reviewed relevance checks for grounded search

Big Data Big Impact, Georgia Tech Jan 2026 - Present
Director of Projects *Atlanta, GA*

- Lead 11 AI/ML and software teams and 100+ students building production data, NLP, and model-driven systems
- Set evaluation plans, data requirements, test gates, and release criteria across projects with different ML stacks
- Drove 6 projects from requirements to deployment and presented technical tradeoffs to GT advisors and Microsoft

Georgia Institute of Technology May 2026 - Aug 2026
Teaching Assistant, CS 2340: Objects and Design *Atlanta, GA*

- Sole TA for a summer section covering Java, OOP, design patterns, Agile/Scrum, and software architecture
- Reviewed class and sequence designs for interface boundaries, testability, and maintainability before implementation
- Guided debugging and refactoring in office hours, helping students turn brittle code into modular, testable designs

Georgia Tech VIP - EMADE Lab (Advisor: Dr. Jason Zutty) Aug 2025 - May 2026
Undergraduate Researcher, Evolutionary AutoML *Atlanta, GA*

- Integrated Gemma, Qwen, Gemini, and DeepSeek-R1 as modular operators in a Python evolutionary search framework
- Engineered an island-migration scheduler across parallel Linux workers, reducing time to convergence by 5.8x
- Ran controlled ablations and system tuning that improved synthesized-solution quality by 14%

Projects

FinLoRA | github.com/AayushP-418/Finlora Python | PyTorch | Hugging Face | QLoRA | vLLM

- Audited 28.5K synthetic SEC QA pairs, found 82% placeholder leakage, and rebuilt the data pipeline before retraining
- Compared base, SFT, and DPO variants using exact match, ROUGE, an LLM judge, and bootstrap confidence intervals
- Served the fine-tuned 3.8B model through FastAPI/vLLM after validating the revised evaluation pipeline

Research Paper RAG | github.com/AayushP-418/research-paper-RAG Python | FastAPI | PostgreSQL | pgvector/HNSW

- Created a RAG service with durable Postgres conversations, document APIs, and cited vector retrieval
- Used schema-constrained extraction and source-grounded retrieval to keep model outputs tied to evidence
- Tracked latency, failures, citations, token use, and cost to isolate retrieval and generation failure modes

Vitalis | github.com/AayushP-418/Vitalis-Med-Assistant Python | React | Gemini | ChromaDB | Tool Calling

- Developed an agentic assistant using Gemini tool calls across multi-step parsing, retrieval, and action workflows
- Persisted memory in ChromaDB with escalation, safe fallback, and human approval before higher-risk actions

TerraTrends | github.com/AayushP-418/TerraTrends Python | LSTM/ARIMA | Flask | GCP

- Integrated 15+ public and cloud data sources into county-level economic forecasting workflows
- Modeled 6 economic indicators with LSTM/ARIMA and exposed forecasts through Flask-backed geospatial views

FeatureFlag | github.com/AayushP-418/Feature-Flag Go | Redis | AWS SQS | Terraform

- Added Redis write-through caching to Go services, raising same-machine throughput 30% in a controlled local benchmark
- Cut config staleness from 59s to under 1s with SQS invalidation and server-sent updates to SDK clients